

Ablepharon-Macrostomia Syndrome (AMS): Comprehensive Disease Characteristics Report

Disease: Ablepharon-Macrostomia Syndrome **MONDO ID:** MONDO:0008693 **OMIM:** #200110
Category: Mendelian (autosomal dominant, de novo) **Causal gene:** *TWIST2* (a.k.a. *DERMO1*), HGNC:20670, chromosome 2q37.3

Summary

Ablepharon-Macrostomia Syndrome (AMS) is an ultra-rare congenital ectodermal dysplasia (fewer than ~20 well-documented cases worldwide since its first description in 1977) defined by a nearly pathognomonic combination of **ablepharon** (absent or severely foreshortened eyelids with anterior-lamellar dysgenesis) and **macrostomia** (an abnormally wide, "fish-shaped" mouth), accompanied by absent eyebrows/eyelashes, external-ear malformations, malar/zygomatic hypoplasia, sparse hair/alopecia, dry ichthyotic redundant skin, rudimentary nipples, and ambiguous or abnormal genitalia. The disorder is caused by a **recurrent de novo heterozygous missense mutation in *TWIST2*, c.223G>A (p.Glu75Lys / E75K)**, located in the basic DNA-binding domain of this basic helix-loop-helix (bHLH) transcription factor ([PMID: 26119818](#)). Remarkably, mutation of the *same* codon to glutamine or alanine (E75Q/E75A) produces the allelic **Barber-Say syndrome (BSS)** rather than AMS, one of the cleanest genotype–phenotype correlations in dysmorphology.

Mechanistically, the E75K substitution does not eliminate the protein (that would cause a different, recessive disease — Setleis syndrome); instead it **alters the DNA-binding specificity** of *TWIST2*, producing a dominant antimorphic/neomorphic effect that dysregulates the Wnt-downstream ***TWIST2/DERMO1* mesenchymal transcriptional program** that governs dermal, craniofacial, and skin-appendage development. The result is faulty specification and differentiation of cranial-neural-crest- and mesoderm-derived mesenchymal progenitors, producing the characteristic eyelid, malar, ear, skin, and genital defects. The disorder is faithfully modeled in **zebrafish** (a base-edited *twist2* p.E78K knock-in) and in *C. elegans* (an allelic series engineered into the single Twist homolog *hlh-8*).

There is **no disease-modifying pharmacologic or gene therapy**. Management is multidisciplinary, supportive, and surgical: **urgent neonatal corneal protection** (intensive lubrication, masquerade flaps) to prevent exposure keratopathy and blindness, followed by **staged anterior-lamellar eyelid reconstruction** (e.g., full-thickness skin grafts over the Müller-muscle/conjunctiva complex, rib-cartilage/fat grafting, reverse hatchet flaps), plus reconstructive surgery for macrostomia, ear, skin, and genital anomalies. Cognition and lifespan are generally normal; the major morbidity is ophthalmic. Both sexes are affected, there is no known population/founder enrichment, and there is no known non-genetic (environmental, infectious, lifestyle) contribution or prevention beyond genetic counseling.

1. Disease Information

Overview. AMS is an extremely rare congenital malformation syndrome of ectoderm- and mesenchyme-derived structures. It is characterized by absent/short eyelids (ablepharon), a wide fish-shaped mouth (macrostomia), external ear abnormalities, absent eyebrows/eyelashes, and a range of skin, skeletal, and genital anomalies ([PMID: 15103726](#)). It was first described by McCarthy and West in 1977.

Key identifiers. | Resource | Identifier | |---|---| | MONDO | MONDO:0008693 | | OMIM | #200110 (Ablepharon-Macrostomia Syndrome) | | Orphanet | ORPHA:920 | | MeSH | Ablepharon macrostomia syndrome (within "Ectodermal Dysplasia" tree) | | Gene | *TWIST2* / *DERMO1*, HGNC:20670, NCBI Gene 117581 | | ICD-10 | Q18.8 / Q10.3 (other congenital malformations of face/eyelid) — no specific code | | ICD-11 | LD2F.1Y (other specified developmental anomalies) — no dedicated code |

Synonyms / alternative names. Ablepharon-macrostomia syndrome; AMS. It is closely related to (and allelic with) **Barber-Say syndrome (BSS)**, and both belong to the *TWIST2*-related craniofacial/ectodermal dysplasia spectrum.

Data provenance. The disease-level knowledge base for AMS is derived almost entirely from **aggregated case reports and small case series** (individual patients described in the published literature) plus functional/molecular studies, rather than from large EHR cohorts or registries. Given fewer than ~20 reported cases, all epidemiologic and clinical statements are based on this aggregated case literature.

2. Etiology

Disease causal factors — genetic. AMS is a monogenic Mendelian disorder. The primary and essentially sole cause is a **de novo heterozygous missense mutation in *TWIST2***, specifically **c.223G>A, p.Glu75Lys (E75K)** in the basic DNA-binding domain ([PMID: 26119818](#); [PMID: 28369379](#)). Inheritance is autosomal dominant; most cases are sporadic (de novo), with rare familial transmission and rare mosaic cases.

Genetic risk factors. The single causal variant is the risk factor. There are no known susceptibility loci or modifier genes established for AMS beyond the allelic-series relationship at codon 75 that determines AMS vs BSS. No GWAS/polygenic contribution applies to this Mendelian condition.

Environmental risk factors. None identified. No toxins, teratogens, infections, parental age effects, occupational exposures, diet, or lifestyle factors have been implicated. The de novo mutations arise spontaneously.

Protective factors. None identified (neither genetic protective alleles nor environmental/dietary protective exposures are described for this single-gene disorder).

Gene–environment interactions. Not applicable / none reported. AMS is a fully penetrant consequence of the causal genotype; there is no evidence of environmental modulation.

3. Phenotypes

AMS phenotypes are **congenital (present at birth), non-progressive/stable** structural malformations (the ophthalmic complications, however, can progress if untreated), and highly **consistent** across reported patients for the two cardinal features. Frequencies below are qualitative given the tiny case literature.

Phenotype	Type	HPO term (suggested)	Onset	Frequency	Notes
Ablepharon / absent or short eyelids (anterior-lamellar dysgenesis)	Physical/structural	HP:0011500 (Ablepharon)	Congenital	Cardinal (~100%; "absent" lids in ~60% of reports)	Sight-threatening; anterior lamella specifically deficient (PMID: 34092176)
Macrostomia (wide fish-shaped mouth)	Physical/structural	HP:0000154 (Macrostomia)	Congenital	Cardinal (~100%)	
Absent eyebrows and eyelashes	Physical	HP:0002223 (Absent eyebrow); HP:0000561 (Absent eyelashes)	Congenital	Very frequent	
External ear abnormalities / dysplastic ears	Physical	HP:0000356 (Abnormal outer ear morphology)	Congenital	Frequent	
Malar/zygomatic hypoplasia; zygomatic-arch absence	Skeletal	HP:0000272 (Malar flattening)	Congenital	Frequent	Zygomatic arch absence reported (PMID: 3293678)
Alopecia / sparse hair; absent lanugo	Ectodermal	HP:0001596 (Alopecia); HP:0008070 (Sparse hair)	Congenital	Frequent	
Dry, ichthyotic, redundant skin	Ectodermal/skin	HP:0008064 (Ichthyosis);	Congenital	Frequent	

Phenotype	Type	HPO term (suggested)	Onset	Frequency	Notes
		HP:0001582 (Redundant skin)			
Rudimentary/ absent nipples	Physical	HP:0002557 (Rudimentary/ absent nipples)	Congenital	Reported	
Ambiguous / abnormal genitalia (e.g., absent prepuce)	Physical	HP:0000078 (Abnormality of the genital system)	Congenital	Frequent in both sexes	Absent prepuce reported (PMID: 34850759)
Hypertelorism	Physical	HP:0000316 (Hypertelorism)	Congenital	Reported (PMID: 33055564)	
Exposure keratopathy / corneal ulceration → vision loss	Ophthalmic complication (secondary)	HP:0000508 / HP:0000481	Neonatal onset, progressive if untreated	Major morbidity	Preventable with early intervention (PMID: 38967579)
Laryngo- tracheal malacia / stenosis	Visceral (rare)	HP:0001601 (Laryngomalacia)	Variable	Rare	First stenosis case: PMID: 31462237

Canonical description (F003): "AMS is characterized by absent or short eyelids, absent eyebrows and eyelashes, macrostomia, and external ear abnormalities. Additional features include alopecia or sparse hair, hypoplastic malar region, redundant skin, rudimentary nipples, abnormal genitalia" ([PMID: 15103726](#)).

Severity/expressivity. Variable; **mosaic TWIST2 expression yields a milder phenotype**, with anterior-lamella abnormality remaining a common feature ([PMID: 34092176](#)).

Quality-of-life impact. The dominant QoL burden is **ophthalmic** — the eyelid defect causes lagophthalmos and exposure keratopathy that, untreated, leads to corneal ulceration and blindness ([PMID: 38967579](#)). Facial dysmorphism (macrostomia, ear, malar) carries

psychosocial and functional (feeding, speech) impact requiring multidisciplinary reconstruction (PMID: 33689605). Cognition is generally normal. A patient's-view account underscores the lived experience of AMS/BSS (PMID: 28690482). No standardized EQ-5D/SF-36 QoL data exist for this ultra-rare disorder.

4. Genetic / Molecular Information

Causal gene. *TWIST2* (also **DERMO1**), a bHLH transcription factor gene on chromosome 2q37.3 (HGNC:20670; OMIM *607556). UniProt Q8WVJ9.

Pathogenic variant. - **AMS-specific variant: c.223G>A, p.Glu75Lys (E75K)** — a **missense** change in the **basic DNA-binding domain** (PMID: 26119818). - **Classification:** Pathogenic (recurrent de novo, robust genotype–phenotype correlation, functional evidence) per ACMG/AMP criteria (PS2, PS1/PM5 at codon, PM1 in critical domain, PP3). - **Allele frequency:** Absent from population databases (gnomAD) — de novo, not a population polymorphism. - **Origin:** **Germline de novo** (germline/somatic mosaicism described in rare milder cases, PMID: 34092176). Not somatic/cancer-associated. - **Functional consequence: Altered DNA binding** — a **dominant antimorphic/neomorphic** effect, *not* simple loss of function or haploinsufficiency. *"All identified mutations fell in the basic domain of TWIST2 and altered the DNA-binding pattern of Flag-TWIST2 in HeLa cells"* (PMID: 26119818); basic-domain substitutions *"exert antimorphic effects"* (PMID: 30450715).

The TWIST2 allelic series (F002) — a key organizing principle:

Variant	Residue	Disease	Inheritance	Mechanism
<i>TWIST2</i> p.Glu75Lys	E75 (basic domain)	AMS (this disease)	AD, de novo	Altered DNA binding (antimorph)
<i>TWIST2</i> p.Glu75Gln / Ala	E75 (basic domain)	Barber-Say syndrome	AD, de novo	Altered DNA binding (antimorph)
<i>TWIST2</i> biallelic loss-of-function	—	Setleis syndrome / FFDD3 (MIM#227260)	AR	Loss of function
<i>TWIST1</i> p.Glu117 (paralogous residue)	basic domain	Sweeney-Cox syndrome	AD	Altered DNA binding
<i>TWIST1</i> haploinsufficiency	—	Saethre-Chotzen syndrome	AD	Loss of function (+ craniosynostosis)

"a lysine at *TWIST2* residue 75 resulted in AMS, whereas a glutamine or alanine yielded BSS" (PMID: 26119818). "subjects with Barber-Say and Ablepharon-Macrostomia syndromes were found to harbor heterozygous missense substitutions in the paralogous glutamic acid residue in *TWIST2* (p.Glu75Ala, p.Glu75Gln and p.Glu75Lys)" and "This allelic series revealed that different substitutions exhibit graded severity, in terms of both gene expression and cellular phenotype" (PMID: 28369379). "Setleis syndrome (SS), or focal facial dermal dysplasia type III (FFDD3, MIM #227260), is an autosomal recessive condition caused by biallelic loss-of-function variants in *TWIST2*" (PMID: 36942595).

Modifier genes. None established. The E→ (K vs Q/A) identity at codon 75 is itself the principal genotype-driven modifier of phenotype (AMS vs BSS).

Epigenetic information. No disease-specific methylation signature is described for AMS. Of note, in *cancer* cell lines the *DERMO1/TWIST2* promoter can be silenced by CpG-island hypermethylation (PMID: 21109964) — relevant to *TWIST2* biology as a tumor-suppressor context but not to AMS pathogenesis.

Chromosomal abnormalities. None cause AMS; karyotype/CMA are normal in AMS and serve mainly to exclude CNV mimics. (By contrast, a 1p36.23p36.22 dosage change relates to the *Setleis/FFDD3* spectrum, PMID: 36942595.)

5. Environmental Information

- **Environmental factors:** None known. AMS is not caused or triggered by toxins, radiation, pollution, or occupational exposure.
- **Lifestyle factors:** Not applicable — no smoking/diet/alcohol association (congenital de novo genetic disorder).
- **Infectious agents:** None. AMS is not infectious in origin and has no known infectious trigger.

This section is essentially "**not applicable**" for AMS: it is a purely genetic, de novo Mendelian disorder.

6. Mechanism / Pathophysiology

Ordered causal chain (initiating lesion → clinical manifestation)

1. A **de novo heterozygous mutation** *TWIST2* c.223G>A (p.Glu75Lys) arises in the germline, placing a lysine in the basic DNA-binding domain of the *TWIST2* (DERMO1) bHLH transcription factor. → *leads to*
2. **Altered DNA-binding specificity** of the mutant *TWIST2* protein (demonstrated: altered binding of Flag-*TWIST2* in HeLa cells), producing a **dominant antimorphic/neomorphic** activity rather than loss of function. → *results in*
3. **Dysregulation of the *TWIST2*/DERMO1 transcriptional program** in mesenchymal progenitor cells. *TWIST2* normally dimerizes with E-proteins, binds E-box DNA, and acts **downstream of canonical Wnt/ β -catenin signaling** as a master regulator of dermal/mesenchymal lineage identity. → *leads to*
4. **Impaired specification and differentiation of cranial-neural-crest- and mesoderm-derived mesenchymal progenitors** (dermal fibroblasts, cranial mesenchyme, chondrogenic/osteogenic and adipogenic precursors). → *results in*
5. **Faulty morphogenesis of ectoderm-associated mesenchymal structures:** eyelid anterior lamella, malar/zygomatic bone, auricular cartilage, hair follicles, skin, nipples, and genitalia. → *manifests as*
6. **Clinical phenotype:** ablepharon, macrostomia, malar/zygomatic hypoplasia, ear dysplasia, sparse hair, redundant/ichthyotic skin, rudimentary nipples, ambiguous genitalia.

7. [Downstream / inferred, secondary] Anterior-lamella deficiency → **lagophthalmos** → **corneal exposure** → **exposure keratopathy** → **corneal ulceration** → **blindness if untreated**.

Branch note: Steps 1–3 are demonstrated *in vitro* and in animal models; steps 4–5 are inferred from TWIST2/DERMO1 developmental biology; step 7 is a well-documented secondary mechanical consequence.

Supporting detail by category

- **Molecular pathways.** Canonical **Wnt/ β -catenin** → **DERMO1/TWIST2** axis. "*Wnt signaling/ β -catenin is absolutely required and sufficient for Dermo1 expression and dermal cell identity in the cranium*" (PMID: 20980404). In zebrafish skin-appendage development, "*the expression of twist2/dermo1 and twist3 is regulated by Wnt signaling*" (PMID: 33994357). GO suggestions: **GO:0016055** (Wnt signaling pathway), **GO:0006357** (regulation of transcription by RNA Pol II), **GO:0003700** (DNA-binding transcription factor activity).
- **Cellular processes.** Mesenchymal stem-cell self-renewal and lineage commitment; suppression of osteogenesis and promotion of adipogenesis via Id genes. "*implicate the TWIST gene family members as potential mediators of MSC self-renewal and lineage commitment in postnatal skeletal tissues*" (PMID: 19609939). GO: **GO:0060485** (mesenchyme development), **GO:0001837** (EMT), **GO:0030154** (cell differentiation).
- **Protein dysfunction.** bHLH transcription factor with altered DNA-binding; **gain-of-abnormal-function/antimorph**, not misfolding/aggregation. Basic-domain substitutions "*exert antimorphic effects*" (PMID: 30450715).
- **Metabolic changes / immune involvement / oxidative injury:** Not primary to AMS. (Twist2-null mice — a *loss-of-function* context distinct from AMS — show elevated proinflammatory cytokines and perinatal death, illustrating TWIST2's role restraining inflammation, but this is not the AMS mechanism.)
- **Biochemical abnormality.** Transcription-factor DNA-binding defect (no enzyme/ion-channel deficiency).
- **Molecular profiling.** Mutant TWIST2 expressed in zebrafish embryos causes **widespread transcriptome changes** and abnormal developmental phenotypes (PMID: 26119818); the *C. elegans hlh-8* allelic series shows graded gene-expression changes (PMID: 28369379). No human patient transcriptomic/proteomic/metabolomic datasets exist.

Cell types (CL suggestions): dermal fibroblast (CL:0000057), mesenchymal stem cell (CL:0000134), cranial neural crest cell (CL:0000333), chondrocyte (CL:0000138). **GO biological processes:** mesenchyme development (GO:0060485), skin development (GO:0043588), palate development, hair follicle morphogenesis (GO:0031069).

7. Anatomical Structures Affected

Organ level (primary). - **Eyelids / anterior lamella** (UBERON:0000014 skin of eyelid; UBERON:0001711 eyelid) — cardinal. - **Mouth / oral opening** (UBERON:0000165 mouth) — macrostomia. - **External ear / auricle** (UBERON:0001757 pinna). - **Malar/zygomatic bone** (UBERON:0001683 zygomatic bone) — hypoplasia/arch absence. - **Skin** (UBERON:0002097) — ichthyotic, redundant. - **Hair follicles** (UBERON:0002073) — sparse hair/alopecia. - **Nipple/breast** (UBERON:0002030) — rudimentary nipples. - **External genitalia** (UBERON:0000990 reproductive system) — ambiguous/abnormal.

Secondary organ involvement. - **Cornea/ocular surface** (UBERON:0000964) — exposure keratopathy, ulceration (mechanical consequence of eyelid defect). - **Larynx/trachea** (UBERON:0001737 / UBERON:0003126) — rare malacia/stenosis ([PMID: 31462237](#)).

Body systems: integumentary (skin/hair/nails), musculoskeletal (craniofacial bone/cartilage), ocular/adnexal, reproductive/genitourinary, and (rarely) respiratory.

Tissue/cell level. Predominantly **mesenchyme-derived connective tissue** (dermis, cranial mesenchyme, cartilage/bone precursors) and their epithelial appendages. Targeted cell populations: **dermal fibroblasts (CL:0000057)**, **mesenchymal stem cells (CL:0000134)**, **cranial neural crest cells (CL:0000333)**, **chondrocytes (CL:0000138)**.

Subcellular level. The molecular lesion resides in the **nucleus** (GO:0005634) — a DNA-binding transcription factor acting on chromatin (GO:0003700 DNA-binding TF activity; GO:0000981).

Localization / lateralization. Craniofacial midline and paired bilateral structures; the eyelid, ear, and malar defects are typically **bilateral** (e.g., "bilateral absence or hypoplasia of lower eyelids," [PMID: 3293678](#)).

8. Temporal Development

- **Onset: Congenital** — all structural features are present at birth; the disorder is fully developmental. Onset pattern is fixed/congenital rather than acute or progressive.
 - **Progression:** The malformations themselves are **stable/non-progressive** (structural). The **ophthalmic complication is time-critical**: exposure keratopathy can develop within the first days of life and progress to corneal ulceration and permanent vision loss without intervention ([PMID: 29538102](#); [PMID: 38967579](#)).
 - **Disease course / duration:** Chronic/lifelong structural condition; managed by staged reconstruction over childhood into adulthood. Patients survive into adulthood (reported at ages 37 and 46 years: [PMID: 31462237](#); [PMID: 15103726](#)).
 - **Critical periods:** The **neonatal period** is the key window of vulnerability (corneal protection) and of intervention opportunity. Staged eyelid reconstruction thereafter shows durable results at 10–15 year follow-up ([PMID: 31373987](#)).
 - **Remission:** Not applicable (structural congenital malformation); surgical correction is the only route to functional improvement.
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9. Inheritance and Population

Epidemiology. Ultra-rare. *"Only 15 patients with AMS have been described in 12 articles"* ([PMID: 31373987](#)); *"fewer than 20 cases being reported in the literature"* ([PMID: 33055564](#)). Prevalence/incidence are not formally quantifiable but far below 1/1,000,000. No population-based registry data exist.

Genetic etiology. - **Inheritance pattern:** Autosomal dominant, almost always **de novo**; rare familial transmission and rare **mosaic** cases ([PMID: 31462237](#); [PMID: 39792429](#); [PMID: 34092176](#)). - **Penetrance:** Complete for the constitutional E75K genotype (mosaicism attenuates severity). - **Expressivity:** Variable; **mosaic expression → milder phenotype** ([PMID: 34092176](#)). - **Anticipation:** Not applicable (not a repeat-expansion disorder). - **Germline/somatic mosaicism:** Reported in rare milder cases. - **Founder effects / consanguinity:** None described; parents typically unrelated ([PMID: 34850759](#)). (Consanguinity is relevant to the *recessive* Setleis syndrome, not AMS.) - **Carrier frequency:** Not applicable (dominant, de novo).

Population demographics. - **Affected populations:** Worldwide, no ethnic enrichment; cases reported across regions including the first West African report ([PMID: 34850759](#)). - **Geographic distribution:** None (sporadic, global). - **Sex ratio:** Both sexes affected; ambiguous genitalia reported in both males and females (no strong sex bias). - **Age distribution:** Diagnosed at birth; patients span neonates to at least the fifth decade.

10. Diagnostics

Diagnostic approach (F008). Diagnosis is **clinical gestalt** — the pathognomonic combination of ablepharon + macrostomia with ectodermal/craniofacial/genital features at birth — **confirmed by molecular testing**.

- **Genetic testing:**
 - **Single-gene *TWIST2* sequencing** for the recurrent **c.223G>A (p.Glu75Lys)** is the confirmatory test of choice; the specific codon-75 substitution also distinguishes AMS (E75K) from allelic BSS (E75Q/E75A) ([PMID: 26119818](#)).
 - **Whole-exome/whole-genome sequencing** is useful when the diagnosis is uncertain or for gene discovery.
 - **Chromosomal microarray / karyotype** are typically **normal**; their role is to exclude CNV/aneuploidy mimics.
 - Mitochondrial DNA, repeat-expansion, and FISH testing are **not indicated**.
 - **Clinical tests / imaging:** Skull radiography/CT can demonstrate **malar/zygomatic hypoplasia or zygomatic-arch absence** ([PMID: 3293678](#); [PMID: 34850759](#)). Ophthalmic slit-lamp examination assesses corneal integrity/exposure. Bronchoscopy is used if airway symptoms suggest laryngo-tracheal malacia/stenosis ([PMID: 31462237](#)).
 - **Laboratory/biomarkers:** No specific biochemical biomarker; diagnosis is molecular. Endocrine/genital work-up may be needed for ambiguous genitalia.
 - **Omics-based diagnostics:** Not used clinically for AMS.

Clinical criteria. No formal consensus diagnostic criteria; diagnosis rests on the characteristic phenotype + *TWIST2* variant.

Differential diagnosis. | Condition | Gene/mechanism | Distinguishing feature | |---|---|---| |
Barber-Say syndrome (allelic) | *TWIST2* E75Q/E75A | **Hypertrichosis** (vs sparse hair in AMS); codon-75 substitution differs ([PMID: 28680619](#)) | | **Saethre-Chotzen syndrome** | *TWIST1*

haploinsufficiency | **Craniosynostosis** — a cardinal feature *absent* in AMS ([PMID: 30450715](#)) | | **Sweeney-Cox syndrome** | *TWIST1* Glu117 | Paralogous *TWIST1* basic-domain substitution ([PMID: 28369379](#)) | | **Setleis syndrome** / **FFDD3** | *TWIST2* biallelic LoF (AR) | Recessive; focal facial dermal dysplasia ([PMID: 36942595](#)) | | **Fraser** / **cryptophthalmos spectrum** | *FRAS1* / *FREM2* / *GRIP1* | Cryptophthalmos (skin covering eye), syndactyly ([PMID: 37493047](#)) |

"craniosynostosis, which is a cardinal feature of Saethre-Chotzen syndrome" — its absence is a key discriminator for AMS ([PMID: 30450715](#)).

Screening. No population newborn screening exists (ultra-rare, clinically obvious at birth). Cascade/carrier screening is generally unnecessary given de novo origin, but molecular confirmation informs recurrence-risk counseling.

11. Outcome / Prognosis

- **Survival / mortality:** Generally **normal life expectancy**; cognition typically normal. AMS is not intrinsically lethal — patients reach adulthood (documented at 37 and 46 years). Rare life-threatening events are secondary (e.g., acute airway compromise from laryngo-tracheal stenosis requiring tracheostomy, [PMID: 31462237](#)).
 - **Morbidity / function:** Dominated by **ophthalmic morbidity** — untreated exposure keratopathy can cause permanent vision loss/blindness ([PMID: 38967579](#); [PMID: 33055564](#)). Facial dysmorphism affects feeding, speech, and psychosocial function.
 - **Disease course / complications:** Exposure keratopathy, corneal ulceration; feeding/airway issues; psychosocial impact. Surgical reconstruction substantially improves function and appearance.
 - **Recovery potential:** Structural defects are **surgically correctable** with good durability — full-thickness skin grafts over Müller muscle gave clear corneas at 10–15 year follow-up ([PMID: 31373987](#)).
 - **Prognostic factors:** **Timeliness of neonatal corneal protection** is the principal determinant of visual outcome. Mosaic cases have milder overall prognosis ([PMID: 34092176](#)).
 - **Prognostic biomarkers:** None molecular; genotype (E75K constitutional vs mosaic) predicts severity.
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12. Treatment

No disease-modifying pharmacotherapy or gene/RNA/cell therapy exists. Management is **supportive, surgical, and multidisciplinary** (F004).

Urgent supportive care (neonatal). Intensive **ocular lubrication** to prevent exposure keratopathy; *"Despite intensive ocular lubrication, severe exposure keratopathy developed within the first days after birth. The eyes were closed using masquerade flaps"* (PMID: 29538102). NCIT suggestions: ocular lubricant therapy, supportive care (NCIT:C15277).

Surgical / interventional — eyelid (anterior-lamella) reconstruction. The eyelids are not truly absent but foreshortened with anterior-lamellar dysgenesis; staged reconstruction is the standard:

Technique	Evidence	Outcome
Full-thickness skin grafts over Müller-muscle/conjunctiva complex	PMID: 31373987	"all 3 cases who underwent upper eyelid lengthening with full thickness skin grafts placed over Müller muscle had clear corneas"; durable at 10–15 yr
Masquerade flaps (urgent), then staged division	PMID: 29538102	Emergency corneal protection in severe neonates
Autologous rib cartilage + fat grafting (lower lid)	PMID: 33055564	First reported use for lower-lid reconstruction
Modified reverse hatchet flap + preputial skin graft	PMID: 38967579	Corneal protection / lid lengthening
Deep skin grafts adjusting eyelid contour/position	PMID: 39792429	Improved eyelid contour and position

Other reconstructive surgery. Macrostomia repair (commissuroplasty), ear reconstruction, malar augmentation, skin and genital surgery, all coordinated multidisciplinarily (PMID: 33689605). Airway stenosis managed by temporary tracheostomy + corticosteroids (PMID: 31462237). NCIT suggestions: reconstructive surgical procedure, skin graft (NCIT:C15325), tracheostomy (NCIT:C51796).

Supportive/rehabilitative. Feeding support, speech therapy, and psychosocial support as needed.

Experimental / advanced therapeutics. None; no ClinicalTrials.gov interventional trials for AMS. Pharmacogenomics, targeted therapy, and immunotherapy are **not applicable**.

Treatment strategy. Algorithm: (1) immediate corneal protection at birth → (2) staged anterior-lamellar eyelid reconstruction → (3) sequential correction of macrostomia, ear, malar, skin, and genital anomalies, timed to growth and function.

13. Prevention

- **Primary prevention:** None possible — de novo mutation; not preventable by risk-factor modification, vaccination, or lifestyle change.
- **Secondary prevention: Early recognition and immediate neonatal corneal protection** is the critical "secondary prevention" that prevents the major preventable complication (blindness) ([PMID: 29538102](#); [PMID: 38967579](#)).
- **Tertiary prevention:** Staged reconstruction and ongoing ophthalmic/multidisciplinary follow-up to prevent complications and disability.
- **Genetic screening / counseling:** For families with an affected child, recurrence risk is low (de novo) but non-zero due to possible **germline mosaicism**; genetic counseling and, where desired, prenatal/preimplantation testing for the known *TWIST2* variant are appropriate. Prenatal ultrasound may detect facial anomalies.
- **Immunization, public-health, environmental, behavioral, prophylactic measures:** Not applicable.

14. Other Species / Natural Disease

- **Taxonomy / natural disease:** No naturally occurring AMS-equivalent disease is reported in companion animals or wildlife (OMIA has no AMS entry). AMS is a human de novo disorder; there is **no zoonotic or cross-species transmission** (not applicable).
- **Orthologous genes:** *TWIST2/DERMO1* is evolutionarily conserved — zebrafish *twist2/dermo1* (and *twist3*), *C. elegans* single Twist homolog ***hlh-8*** (residue Glu29), and mouse *Twist2/Dermo1*. The disease-critical glutamic-acid residue (human Glu75) is conserved across paralogs and orthologs ([PMID: 28369379](#)).

- **Comparative biology:** The graded-severity allelic series across TWIST1/TWIST2 basic-domain substitutions is conserved and reproducible in *C. elegans*, demonstrating deep evolutionary conservation of the Twist-family transcription-factor mechanism (PMID: 28369379).

15. Model Organisms

AMS is well-modeled experimentally (F006):

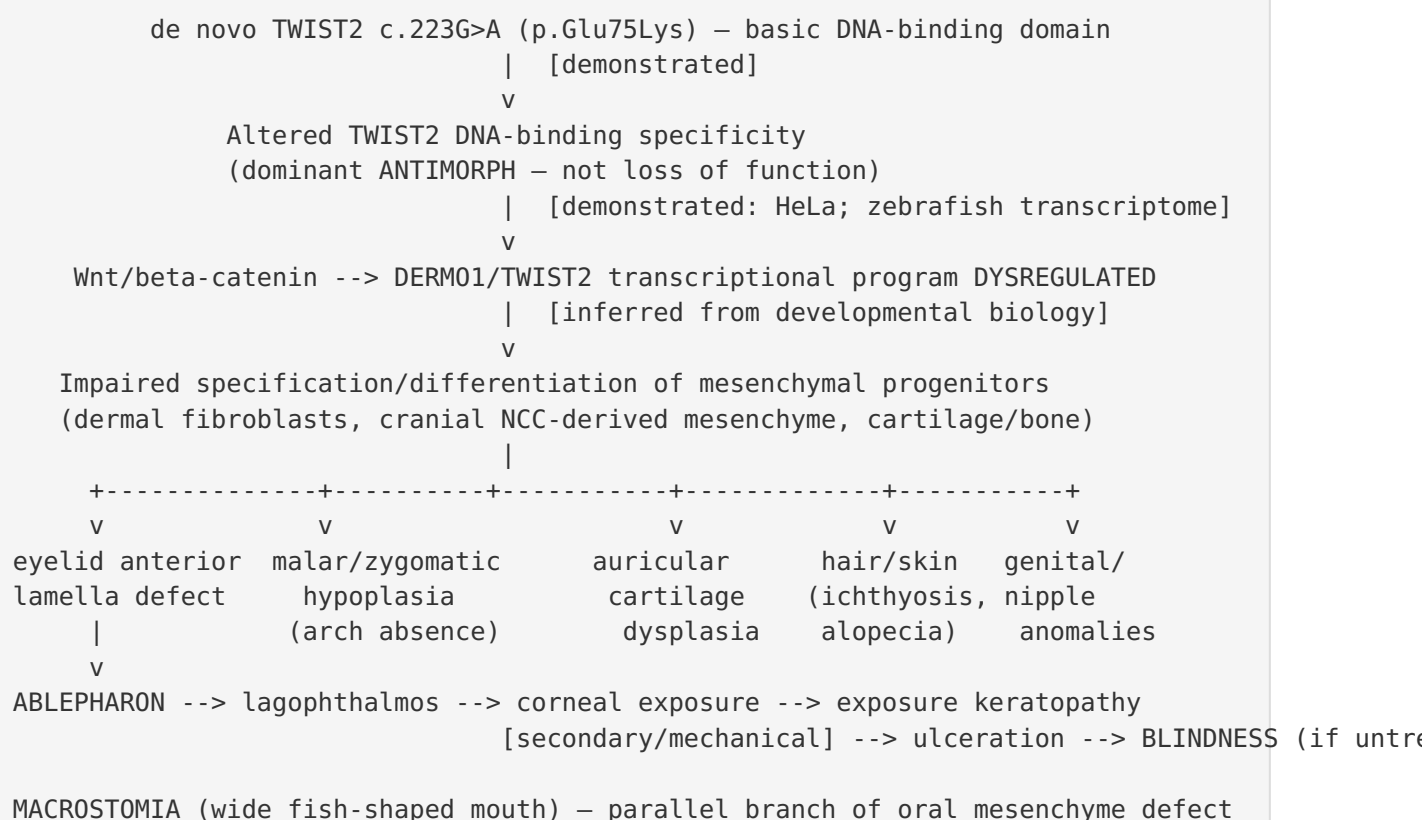
Model	Construct	Recapitulation	Reference
Zebrafish (<i>Danio rerio</i>)	CRISPR base-edited twist2 p.E78K knock-in (paralogous to human E75K)	"recapitulating pathological features of human <i>ablepharon macrostomia syndrome</i> (AMS)"	PMID: 33272268; protocol PMID: 30076894
Zebrafish (overexpression)	Mutant TWIST2 mRNA in embryos	Abnormal developmental phenotypes + widespread transcriptome changes	PMID: 26119818
<i>C. elegans</i>	All five TWIST1/TWIST2 disease alleles engineered into <i>hlh-8</i> Glu29	Graded severity of gene-expression and M-lineage/muscle cellular phenotypes	PMID: 28369379
In vitro (HeLa)	Flag-TWIST2 mutants	Altered DNA-binding pattern	PMID: 26119818
Mouse (<i>Twist2</i> / <i>Dermo1</i>)	Knockout / <i>Dermo1</i> -Cre lineage tools	Developmental biology of Wnt/ β -catenin– <i>Dermo1</i> mesenchyme; <i>Twist2</i> -null → perinatal death with elevated proinflammatory cytokines	PMID: 20980404, PMID: 18231602

"zAncBE4max successfully generated the *Twist2* p.E78K mutation in zebrafish, recapitulating pathological features of human *ablepharon macrostomia syndrome* (AMS)" (PMID: 33272268).

"we engineered all five disease-associated alleles into the equivalent Glu29 residue encoded by *hlh-8*, the single *Twist* homolog present in *Caenorhabditis elegans*" (PMID: 28369379).

Applications: These models enable study of the antimorphic DNA-binding mechanism, the allelic-series severity gradient (AMS vs BSS vs Sweeney-Cox), and downstream transcriptional dysregulation. **Limitations:** Invertebrate/fish models incompletely capture human craniofacial/ectodermal complexity; no mouse knock-in of the exact E75K antimorph is reported for full AMS phenotype recapitulation. **Resources:** ZFIN (zebrafish), WormBase (*C. elegans*), MGI (mouse *Twist2*).

Mechanistic Model / Interpretation



Key interpretive points. (1) The **allele identity at codon 75** determines the disease (K→AMS, Q/A→BSS), while the **mechanism** determines the axis (dominant antimorph → AMS/BSS; recessive LoF → Setleis) — a textbook demonstration of how different mutation *types* in one gene yield distinct diseases. (2) The mechanism is a **transcription-factor DNA-binding perturbation upstream of a broad developmental program**, explaining the pleiotropic,

multi-structure phenotype. (3) The most clinically actionable step is **downstream and mechanical** (eyelid → cornea), which is why timely surgical/supportive intervention — not molecular therapy — currently drives outcomes.

Evidence Base

PMID	Contribution	Supports
26119818	Landmark: recurrent basic-domain <i>TWIST2</i> mutations; E75K→AMS, E75Q/A→BSS; altered DNA binding in HeLa; zebrafish transcriptome	F001, F002, F007, F008 (core causal + mechanism)
28369379	<i>TWIST1/TWIST2</i> allelic series; <i>C. elegans hlh-8</i> modeling; graded severity	F002, F006
36942595	Recessive LoF <i>TWIST2</i> → Setleis/FFDD3 (contrasts dominant AMS)	F002, F008
30450715	Antimorphic effect of basic-domain substitutions; craniosynostosis distinguishes Saethre-Chotzen from AMS	F002, F005, F007, F008
15103726	Canonical clinical feature list (46-yr-old patient)	F003
3293678	Zygomatic-arch absence; bilateral lid involvement	F003, F007
34092176	Mosaic <i>TWIST2</i> → milder phenotype; anterior-lamella hallmark	F003, F005
34850759	First West African case; absent prepuce; unrelated parents	F003, F005
31462237	Laryngo-tracheal stenosis; AD mutation; adult survival	F003, F005
38967579	Sight-threatening keratopathy; reverse hatchet flap	F003, F004
31373987	Definitive skin-graft-over-Müller technique; 10–15 yr outcomes; "only 15 patients"	F004, F005
29538102	Masquerade flap; urgent neonatal corneal protection	F004
33055564	Rib-cartilage/fat grafting; "<20 cases"	F004, F005
39792429	Deep skin grafts; AD inheritance statement	F004, F005
33689605	Multidisciplinary care	F004
33272268	Zebrafish <i>twist2</i> E78K knock-in recapitulates AMS	F006
30076894	Base-editing protocol reproducing the AMS mutation in zebrafish	F006
20980404	Wnt/ β -catenin required/sufficient for Dermo1 + cranial dermal identity	F007

PMID	Contribution	Supports
33994357	twist2/dermo1 Wnt-regulated in skin-appendage development	F007
19609939	TWIST family mediates MSC self-renewal/lineage commitment	F007
28680619	Barber-Say (hypertrichosis) differential	F008
37493047	Cryptophthalmos/Fraser differential	F008
28690482	Patient's-view QoL perspective	Section 3
21109964	DERMO1 promoter methylation (cancer context)	Section 4 (epigenetics, non-AMS)
18231602	Dermo1-Cre/ β -catenin mesenchymal lineage biology	Section 6/15

Evidence-type mix: human clinical (case reports/series), in vitro (HeLa DNA-binding), and model organism (zebrafish, *C. elegans*, mouse). There are **no** large human -omics cohort datasets for AMS.

Limitations and Knowledge Gaps

1. **Tiny evidence base.** Fewer than ~20 reported patients; all clinical/epidemiologic claims rest on aggregated case reports — no registries, no prevalence/incidence figures, no controlled QoL data.
2. **Mechanism partly inferred.** Steps 1–3 (mutation → altered DNA binding → transcriptome change) are demonstrated; steps 4–5 (tissue-specific developmental failure) are inferred from TWIST2/DERMO1 biology, not directly from AMS patient tissue.
3. **No exact mammalian knock-in.** The zebrafish E78K and *C. elegans* *hlh-8* models are informative but phylogenetically distant; no mouse *Twist2* E75K knock-in recapitulating the full craniofacial AMS phenotype has been reported.
4. **No molecular therapy.** No pharmacologic, gene-editing, or ASO approach is in development; the antimorphic/dominant mechanism (rather than simple LoF) complicates gene-replacement strategies.

5. **Genotype–phenotype nuance unresolved.** Why lysine (E75K) specifically yields AMS while glutamine/alanine yield BSS — i.e., the precise altered DNA-binding readout — is not fully mapped at the level of target genes.
 6. **Under-characterized systemic involvement.** Rare visceral features (airway, genital) are described anecdotally; their true frequency and natural history are unknown.
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Proposed Follow-up Experiments / Actions

1. **Patient-derived iPSC / organoid models:** Generate *TWIST2* E75K iPSCs and differentiate into cranial-neural-crest and dermal mesenchyme to directly map the dysregulated transcriptional program (RNA-seq + CUT&RUN/ChIP-seq for mutant vs WT *TWIST2* genome-wide binding).
 2. **Precise mouse knock-in:** Create a conditional *Twist2* E75K knock-in mouse to test craniofacial/eyelid/skin phenotype recapitulation and to serve as a preclinical platform.
 3. **Define the altered-binding "neo-target" set:** Comparative genome-wide binding of E75K vs E75Q/A vs WT to explain the AMS-vs-BSS divergence mechanistically.
 4. **Allele-selective silencing proof-of-concept:** Given the dominant antimorph, test allele-specific ASO/siRNA or base-editing correction in the zebrafish E78K model as a therapeutic feasibility study.
 5. **International registry + natural-history study:** Pool the ultra-rare cases to quantify feature frequencies, ophthalmic outcomes, and standardized QoL (validated instruments), and to formalize diagnostic/management guidelines.
 6. **Standardize surgical outcome reporting:** Prospective, multi-center comparison of eyelid-reconstruction techniques (skin-graft-over-Müller vs rib-cartilage/fat vs flaps) with corneal and cosmetic endpoints at ≥ 10 years.
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Report compiled from 8 confirmed findings and 29 reviewed papers over a 5-iteration autonomous investigation. Evidence types: human clinical case literature, in vitro DNA-binding assays, and zebrafish/C. elegans/mouse model-organism studies.
